Lagrange–Newton and Sequential Quadratic Programming

The methods of Chapter 15 solve a constrained problem through unconstrained subproblems. We now ask how to retain the constraints in a simpler local model and correct the primal variables and multipliers together. We will first obtain such a correction by applying Newton's method to the equality KKT equations. We will then reinterpret the same equations as the optimality conditions of a quadratic subproblem. This optimization interpretation makes it possible to modify curvature, control step length, and extend the construction to inequality constraints.

Learning objectives

By the end of this chapter, you should be able to:

- derive and implement a Lagrange–Newton (LN) step from the KKT residual;
- state the regularity assumptions behind its local quadratic convergence;
- distinguish a nonsingular KKT system from a well-posed quadratic minimization problem;
- derive sequential quadratic programming (SQP) by changing the multiplier variable;
- explain the Lagrangian Hessian, merit function, and damped SQP update; and
- compare the requirements of LN/SQP with those of QP and ALM.

16.1 From unconstrained subproblems to constrained local models

Let us first place the new methods in the two-level framework of Chapter 15. An outer iteration updates parameters or variables and constructs a subproblem. An inner solver then approximately solves that subproblem. The stopping test concerns the original problem: we seek a primal–multiplier pair whose KKT violation is at most ε_{out}, rather than merely a small subproblem residual. The inner tolerance $\varepsilon_{\text{in}}^{(k)}$ measures the latter and must be chosen with the desired outer accuracy in mind.

For an equality constraint $\boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}$, the two constructions already studied use

$$P_{\rho^{(k)}}(\boldsymbol{x}) = f(\boldsymbol{x}) + \frac{\rho^{(k)}}{2}\,\|\boldsymbol{g}(\boldsymbol{x})\|^2\,, \qquad \mathcal{L}_{\rho}(\boldsymbol{x}, \boldsymbol{\lambda}^{(k)}) = f(\boldsymbol{x}) + (\boldsymbol{\lambda}^{(k)})^{\mathsf{T}}\boldsymbol{g}(\boldsymbol{x}) + \frac{\rho}{2}\,\|\boldsymbol{g}(\boldsymbol{x})\|^2\,.$$

Both inner problems are unconstrained. Their Hessians contain penalty curvature of the form $\rho \boldsymbol{A}\boldsymbol{A}^{\mathsf{T}}$, as well as the appropriate Lagrangian curvature, so raising ρ can make the

inner solve harder. Consequently outer progress, inner error, and penalty selection must be considered together. We now investigate a different subproblem: retain linearized constraints and replace nonlinear optimality equations by a linear system. LN and SQP are again methods constructed from the primal problem, although they also maintain multipliers.

16.2 The Lagrange–Newton method

16.2.1 From the Lagrangian to a system of equations

We retain the equality-constrained model of Equation (15.1):

$$\min_{\boldsymbol{x}\in\mathcal{D}} f(\boldsymbol{x}) \qquad \text{subject to} \qquad \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}, \tag{16.1}$$

where $\mathcal{D} \subseteq \mathbb{R}^n$ is open and $f, g_1, \dots, g_m$ are twice continuously differentiable. Write

$$\boldsymbol{A}(\boldsymbol{x}) = [\nabla g_1(\boldsymbol{x}), \dots, \nabla g_m(\boldsymbol{x})] \in \mathbb{R}^{n\times m}, \qquad L(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^{\mathsf{T}} \boldsymbol{g}(\boldsymbol{x}).$$

Our sign and matrix conventions will be used throughout: equality multipliers have no sign restrictions, and the constraint Jacobian is $\boldsymbol{A}^{\mathsf{T}}$.

To construct a Newton iteration, collect the two KKT equations from Equations (12.7) and (12.8) into a residual with $\boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{\lambda})$:

$$F(\boldsymbol{z}) = \begin{pmatrix} \nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}) \\ \boldsymbol{g}(\boldsymbol{x}) \end{pmatrix}. \tag{16.2}$$

Thus solving the equality KKT system amounts to finding a zero of F. This is a first-order target; a zero of F need not be a minimum in a nonconvex problem. The optimality conclusions still come from Chapters 13 and 14.

We next differentiate the residual block by block. Since

$$\nabla_{\boldsymbol{x}} L = \nabla f + \sum_{i=1}^{m} \lambda_i \nabla g_i, \qquad \boldsymbol{H}_L(\boldsymbol{x}, \boldsymbol{\lambda}) = \nabla^2_{\boldsymbol{x}\boldsymbol{x}} L = \boldsymbol{H}_f + \sum_{i=1}^{m} \lambda_i \boldsymbol{H}_{g_i},$$

the derivative of its first block with respect to $\boldsymbol{x}$ is $\boldsymbol{H}_L$, and with respect to $\boldsymbol{\lambda}$ it is $\boldsymbol{A}$. The second block has $\boldsymbol{x}$-derivative $\boldsymbol{A}^{\mathsf{T}}$ and no multiplier dependence. Hence

$$J_F(\boldsymbol{x}, \boldsymbol{\lambda}) = \begin{pmatrix} \boldsymbol{H}_L(\boldsymbol{x}, \boldsymbol{\lambda}) & \boldsymbol{A}(\boldsymbol{x}) \\ \boldsymbol{A}(\boldsymbol{x})^{\mathsf{T}} & \boldsymbol{0} \end{pmatrix}. \tag{16.3}$$

We will examine the invertibility and curvature of this matrix after deriving the iteration.

For numerical stopping we use the same infinity norms as in Chapter 15:

$$\begin{aligned} r_{\mathrm{stat}}(\boldsymbol{x}, \boldsymbol{\lambda}) &= \|\nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda})\|_\infty, \\ r_{\mathrm{feas}}(\boldsymbol{x}) &= \|\boldsymbol{g}(\boldsymbol{x})\|_\infty, \qquad r_{\mathrm{KKT}} = \max\{r_{\mathrm{stat}}, r_{\mathrm{feas}}\}. \end{aligned} \tag{16.4}$$

We stop when $r_{\mathrm{KKT}} \le \varepsilon_{\mathrm{out}}$. Separate reporting of both blocks helps identify poor constraint scaling or an inaccurate multiplier. Unsubscripted norms in the convergence arguments below are Euclidean norms.

16.2.2 Linearizing stationarity and feasibility together

At $\boldsymbol{z}^{(k)}$, approximate F by its affine Taylor model and set the model to zero:

$$F(\boldsymbol{z}^{(k)}) + J_F(\boldsymbol{z}^{(k)})\Delta\boldsymbol{z}^{(k)} = \boldsymbol{0}.$$

Writing $\Delta\boldsymbol{z}^{(k)} = (\boldsymbol{d}^{(k)}, \boldsymbol{\delta}^{(k)})$ gives

$$\begin{pmatrix} \boldsymbol{H}_L^{(k)} & \boldsymbol{A}^{(k)} \\ (\boldsymbol{A}^{(k)})^{\mathsf{T}} & \boldsymbol{0} \end{pmatrix} \begin{pmatrix} \boldsymbol{d}^{(k)} \\ \boldsymbol{\delta}^{(k)} \end{pmatrix} = - \begin{pmatrix} \nabla_{\boldsymbol{x}} L^{(k)} \\ \boldsymbol{g}^{(k)} \end{pmatrix}. \tag{16.5}$$

The KKT equations come from the *Lagrangian*, and this correction is a *Newton* step for those equations. This gives the name Lagrange–Newton (LN). The pure LN method then takes

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \boldsymbol{d}^{(k)}, \qquad \boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \boldsymbol{\delta}^{(k)}. \tag{16.6}$$

Here $\boldsymbol{\delta}^{(k)}$ is a multiplier *increment*. In particular, replacing $\boldsymbol{\lambda}^{(k+1)}$ by $\boldsymbol{\delta}^{(k)}$ would discard the current multiplier and define a different iteration.

To see what the second block accomplishes, apply Taylor's theorem to each constraint along a trial step $\alpha\boldsymbol{d}$. With the iteration index suppressed, $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{d} = -\boldsymbol{g}$ gives

$$\boldsymbol{g}(\boldsymbol{x} + \alpha\boldsymbol{d}) = (1 - \alpha)\boldsymbol{g}(\boldsymbol{x}) + O(\alpha^2 \|\boldsymbol{d}\|^2). \tag{16.7}$$

For a full step, the constant and linear terms cancel, leaving $\boldsymbol{g}(\boldsymbol{x} + \boldsymbol{d}) = O(\|\boldsymbol{d}\|^2)$. This error is small when $\boldsymbol{d}$ is small, but it is generally nonzero. If the constraints are affine, the remainder vanishes and a full step is exactly feasible. Stationarity is also only linearized; it generally remains nonzero after one update. An initially feasible point is not required: at an infeasible iterate the linearized equation aims to correct its existing violation as well.

Algorithm: Pure Lagrange–Newton

Choose $(\boldsymbol{x}^{(0)}, \boldsymbol{\lambda}^{(0)})$, $\varepsilon_{\text{out}} > 0$, and an update limit. At each iterate:

1. Evaluate both residuals in Equation (16.4); return if their maximum is at most ε_{out}.
2. Form $\boldsymbol{H}_L^{(k)}$, $\boldsymbol{A}^{(k)}$, and solve Equation (16.5).
3. Check that the new primal point lies in $\mathcal{D}$, then take Equation (16.6).

In the supplied file `l16_lagrange_newton_sqp.py`, the function `lagrange_newton` implements this full-step iteration. It solves for $(\boldsymbol{d}, \boldsymbol{\delta})$ and adds both increments. There is no line search in this method.

16.2.3 Local convergence of the full-step iteration

We next examine what happens near a KKT pair. Adapt the Newton argument in Theorem 9.1 to the residual F. Assume that J_F is Lipschitz continuous near a root $\boldsymbol{z}^*$, and that $J_F(\boldsymbol{z}^*)$ is nonsingular. The following estimate verifies that the same quadratic rate applies to the primal–multiplier pair.

Verification of the Newton error estimate. In a sufficiently small convex neighbourhood, let M bound $\|J_F(\boldsymbol{z})^{-1}\|$ and let K be a Lipschitz constant. For $\boldsymbol{e} = \boldsymbol{z} - \boldsymbol{z}^*$, the integral Taylor

formula and $F(\boldsymbol{z}^*) = \mathbf{0}$ give

$$F(\boldsymbol{z}) = \int_0^1 J_F(\boldsymbol{z}^* + t\boldsymbol{e})\boldsymbol{e}\, dt.$$

One Newton step therefore satisfies

$$\boldsymbol{z}^+ - \boldsymbol{z}^* = J_F(\boldsymbol{z})^{-1} \int_0^1 [J_F(\boldsymbol{z}) - J_F(\boldsymbol{z}^* + t\boldsymbol{e})]\boldsymbol{e}\, dt,$$
$$\left\|\boldsymbol{z}^+ - \boldsymbol{z}^*\right\| \le M \int_0^1 K(1-t)\left\|\boldsymbol{e}\right\|^2 dt = \frac{MK}{2}\left\|\boldsymbol{z} - \boldsymbol{z}^*\right\|^2.$$

Shrinking the initial neighbourhood keeps the iterates inside it. Thus full-step LN converges locally quadratically under these assumptions. □

This reasoning requires Lipschitz continuity of the KKT Jacobian, not merely existence of second derivatives. It provides no guarantee that remote full steps decrease either residual or remain in the domain.

16.2.4 A common test: the first step and its trajectory

To examine both the step geometry and computational work, use

$$\min_{\boldsymbol{x}\in\mathbb{R}^2} \tfrac{1}{2}[(x_1-1)^2 + (x_2-2)^2] \quad \text{subject to} \quad g(\boldsymbol{x}) = x_1^2 + x_2 - 1 = 0, \tag{16.8}$$

starting from $\boldsymbol{x}^{(0)} = (1/2, 3/4)^\mathsf{T}$, $\lambda^{(0)} = 0$. For this problem,

$$\nabla f = \begin{pmatrix} x_1 - 1 \\ x_2 - 2 \end{pmatrix}, \quad \boldsymbol{A} = \begin{pmatrix} 2x_1 \\ 1 \end{pmatrix}, \quad \boldsymbol{H}_L = \begin{pmatrix} 1 + 2\lambda & 0 \\ 0 & 1 \end{pmatrix}.$$

At the initial pair the LN system is

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ \delta \end{pmatrix} = \begin{pmatrix} 1/2 \\ 5/4 \\ 0 \end{pmatrix}.$$

Solving gives $\boldsymbol{d} = (-3/8, 3/8)^\mathsf{T}$ and $\delta = 7/8$, hence

$$\boldsymbol{x}^{(1)} = (1/8, 9/8)^\mathsf{T}, \quad \lambda^{(1)} = 7/8, \qquad g(\boldsymbol{x}^{(1)}) = 9/64.$$

Although the initial point is feasible and $d_1 + d_2 = 0$, the full step leaves the parabola. More generally, direct expansion gives $g(\boldsymbol{x}^{(0)} + \alpha\boldsymbol{d}) = 9\alpha^2/64$: the linearized constraint vanishes along the tangent line, while the actual constraint has a quadratic error.

Continuing with full steps gives the following history; for the convergence of KKT violation and the trajectory, see Figure 16.1. Both columns of residuals use the infinity norm in Equation (16.4).

k	$x_1^{(k)}$	$x_2^{(k)}$	$\lambda^{(k)}$	r_{stat}	r_{feas}
0	0.500000	0.750000	0.000000	1.250	0
1	0.125000	1.125000	0.875000	6.563×10^{-1}	1.406×10^{-1}
2	0.345833	0.929167	1.070833	8.649×10^{-2}	4.877×10^{-2}
3	0.312623	0.903370	1.096630	1.713×10^{-3}	1.103×10^{-3}
4	0.312909	0.902088	1.097912	7.322×10^{-7}	8.160×10^{-8}

The fourth update already satisfies $\varepsilon_{\text{out}} = 10^{-6}$. To check the limiting point independently, eliminate $x_2 = 1 - x_1^2$. The restricted objective has derivative $2x_1^3 + 3x_1 - 1$ and second derivative $6x_1^2 + 3 > 0$. It has a unique global minimizer, giving $\boldsymbol{x}^* \approx (0.312908, 0.902088)^{\mathsf{T}}$ and $\lambda^* = 2 - x_2^* \approx 1.097912$.

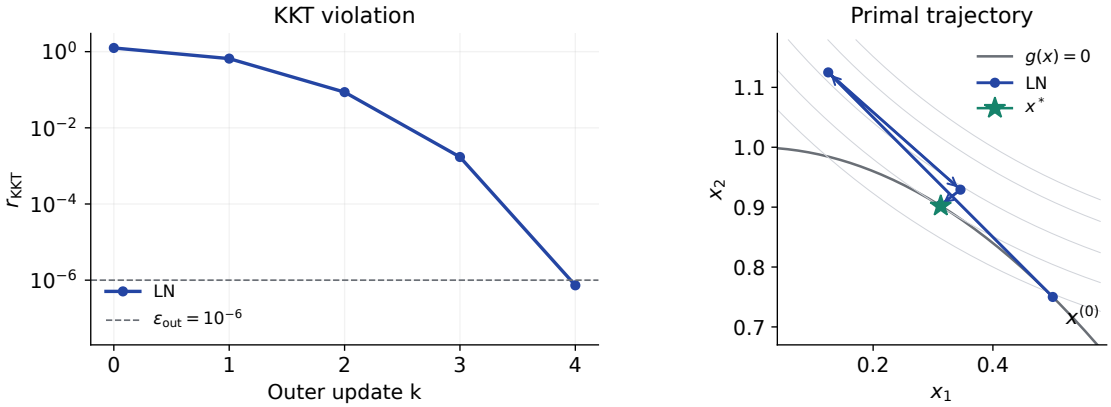


Figure 16.1: Pure LN on Equation (16.8). The KKT violation uses both stationarity and feasibility. The trajectory shows that intermediate iterates need not lie on the feasible parabola.

16.3 Regularity and the limitations of LN

16.3.1 When the Newton system is nonsingular

The local rate argument leaves us with a concrete question: when is the KKT Jacobian invertible? We will reduce the question to the curvature on the linearized tangent space.

Lemma 16.1 (Nonsingularity of the KKT matrix)**.** *Let $\boldsymbol{H}_L \in \mathbb{R}^{n\times n}$ be symmetric, let $\boldsymbol{A} \in \mathbb{R}^{n\times m}$ have full column rank, and let the columns of $\boldsymbol{Z}$ form a basis of* $\operatorname{Null}(\boldsymbol{A}^{\mathsf{T}})$*. Then*

$$\begin{pmatrix} \boldsymbol{H}_L & \boldsymbol{A} \\ \boldsymbol{A}^{\mathsf{T}} & \boldsymbol{0} \end{pmatrix} \text{ nonsingular} \iff \boldsymbol{Z}^{\mathsf{T}}\boldsymbol{H}_L\boldsymbol{Z} \text{ nonsingular.} \tag{16.9}$$

When $m = n$, the tangent space is zero-dimensional and the KKT matrix is always nonsingular.

Proof. A null vector $(\boldsymbol{u}, \boldsymbol{v})$ of the KKT matrix satisfies

$$\boldsymbol{H}_L\boldsymbol{u} + \boldsymbol{A}\boldsymbol{v} = \boldsymbol{0}, \qquad \boldsymbol{A}^{\mathsf{T}}\boldsymbol{u} = \boldsymbol{0}.$$

The second equation gives $\boldsymbol{u} = \boldsymbol{Z}\boldsymbol{w}$. Premultiply the first by $\boldsymbol{Z}^{\mathsf{T}}$, using $\boldsymbol{Z}^{\mathsf{T}}\boldsymbol{A} = \boldsymbol{0}$, to obtain

$$(\boldsymbol{Z}^{\mathsf{T}}\boldsymbol{H}_L\boldsymbol{Z})\boldsymbol{w} = \boldsymbol{0}.$$

If the reduced Hessian is nonsingular, then $\boldsymbol{w} = \boldsymbol{0}$ and hence $\boldsymbol{u} = \boldsymbol{0}$. The first block now gives $\boldsymbol{A}\boldsymbol{v} = \boldsymbol{0}$; full column rank forces $\boldsymbol{v} = \boldsymbol{0}$. Thus the KKT matrix is nonsingular.

For the reverse direction, suppose the reduced Hessian is singular. Choose a nonzero $\boldsymbol{w}$ in its nullspace and set $\boldsymbol{u} = \boldsymbol{Z}\boldsymbol{w} \neq \boldsymbol{0}$. Then $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{u} = \boldsymbol{0}$ and $\boldsymbol{Z}^{\mathsf{T}}\boldsymbol{H}_L\boldsymbol{u} = \boldsymbol{0}$. Since $\operatorname{Null}(\boldsymbol{Z}^{\mathsf{T}}) =$

Range($\boldsymbol{A}$), there is a $\boldsymbol{v}$ such that $\boldsymbol{H}_L\boldsymbol{u} = -\boldsymbol{A}\boldsymbol{v}$. The pair $(\boldsymbol{u}, \boldsymbol{v})$ is a nonzero null vector of the KKT matrix, proving the reverse implication. When $m = n$, the tangent space is $\{\mathbf{0}\}$; there is no reduced system, and the two block equations directly force $\boldsymbol{u} = \boldsymbol{v} = \mathbf{0}$. □

At a KKT pair, LICQ and the equality SOSC from Theorem 13.11 make the reduced Hessian positive definite, hence nonsingular. They therefore supply sufficient conditions for the Newton argument, and nonsingularity persists in a sufficiently small neighbourhood by continuity. Requiring $\boldsymbol{H}_L \succ 0$ is stronger: it imposes positive curvature on all directions, including those that leave the linearized constraints.

To decide whether SOSC is also necessary, first suppose the KKT pair corresponds to a local minimizer. Under LICQ, the SONC gives $\boldsymbol{Z}^{\mathsf{T}}\boldsymbol{H}_L\boldsymbol{Z} \succeq 0$. A symmetric positive semidefinite matrix is nonsingular exactly when it is positive definite. Thus *at a local minimizer under LICQ*, nonsingularity is equivalent to the equality SOSC. At an arbitrary KKT pair, Equation (16.9) only tests for zero eigenvalues; it imposes no sign on the nonzero eigenvalues. The following examples separate these possibilities.

16.3.2 Singularity can occur at a constrained minimum

Consider the smooth problem

$$\min_{(x_1,x_2)\in\mathbb{R}^2} \tfrac{1}{4}x_1^4 + \tfrac{1}{2}x_2^2 \qquad \text{subject to} \qquad x_2 = 0.$$

The unique minimizer is $(0, 0)$, with multiplier zero. Since $\boldsymbol{A} = (0, 1)^{\mathsf{T}}$, LICQ holds. At this pair,

$$\boldsymbol{H}_L = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad J_F = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The tangent direction $(1, 0)^{\mathsf{T}}$ has zero quadratic curvature: the objective grows only to fourth order along the feasible set. The Newton matrix is therefore singular even at the unique minimum. This example explains why existence and uniqueness of a minimizer are not substitutes for the assumptions in the local Newton analysis.

For comparison, replace the objective by $-\frac{1}{2}x_1^2 + \frac{1}{2}x_2^2$. At the origin the reduced Hessian is -1 and the KKT matrix is nonsingular, but the feasible objective is $-x_1^2/2$. The origin is a constrained maximum. Thus an invertible Newton matrix alone gives a root-finding property, not an optimality certificate.

16.3.3 Remote steps need not approach a solution

The local error estimate uses a neighbourhood in which the inverse Jacobian and the Taylor remainder are controlled. To see why such a restriction matters, consider an equality-constrained problem whose tangent component reduces to a scalar Newton iteration:

$$\min_{(u,v)\in\mathbb{R}^2} \tfrac{1}{4}u^4 - u^2 + 2u + \tfrac{1}{2}v^2 \qquad \text{subject to} \qquad v = 0.$$

The KKT equations are $u^3 - 2u + 2 = 0$, $v + \lambda = 0$, and $v = 0$. Starting with $v = \lambda = 0$, LN keeps these two variables zero and updates

$$u^+ = u - \frac{u^3 - 2u + 2}{3u^2 - 2}.$$

From $u^{(0)} = 0$ we obtain $u^{(1)} = 1$, then $u^{(2)} = 0$. Both encountered Newton matrices are nonsingular, but the iteration cycles forever and its stationarity violation alternates between 2 and 1. The quartic objective is coercive and has a global minimum; the cycle is an algorithmic failure rather than nonexistence of a solution.

16.3.4 Why descent on the Lagrangian is not a remedy

To try to safeguard LN, one might imitate unconstrained Newton minimization and line-search directly on L. Let us first examine what function LN is differentiating. For $\boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{\lambda})$,

$$\nabla_{\boldsymbol{z}} L(\boldsymbol{z}) = \begin{pmatrix} \nabla_{\boldsymbol{x}} L \\ \boldsymbol{g} \end{pmatrix} = F(\boldsymbol{z}), \qquad \nabla^2_{\boldsymbol{z}\boldsymbol{z}} L(\boldsymbol{z}) = \begin{pmatrix} \boldsymbol{H}_L & \boldsymbol{A} \\ \boldsymbol{A}^{\mathsf{T}} & \boldsymbol{0} \end{pmatrix} = J_F(\boldsymbol{z}).$$

Thus LN does have the algebraic form of a Newton step for stationary points of L in all variables. But a constrained minimizer does not generally minimize this function jointly in $\boldsymbol{x}$ and $\boldsymbol{\lambda}$.

For example, $\min_x x^2/2$ subject to $x = 0$ has the unique solution and multiplier $(x^*, \lambda^*) = (0, 0)$. Its Lagrangian and Hessian are

$$L(x, \lambda) = \tfrac{1}{2}x^2 + \lambda x, \qquad \nabla^2 L = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

The determinant is -1, so the Hessian is nonsingular and indefinite. Along $\lambda = 0$, $L = x^2/2 > 0$ for $x \neq 0$; along $\lambda = -x$, $L = -x^2/2 < 0$. The KKT pair is a saddle point. More directly, starting from $(x, \lambda) = (1, -1)$, one LN step reaches $(0, 0)$, yet L increases from $-1/2$ to 0. A rule demanding Lagrangian decrease would reject this successful step.

The zero multiplier–multiplier block and the coupling through $\boldsymbol{A}$ create this saddle geometry. It is connected to the dual interpretation of multipliers, which we will develop later. Shifting the entire KKT matrix to positive definiteness and then minimizing L does not resolve the mismatch: it seeks descent toward a joint minimum that the desired KKT pair need not be. We instead turn to an optimization model in the primal step and a merit function that measures both objective progress and feasibility.

16.4 From LN to sequential quadratic programming

16.4.1 Replacing the multiplier increment by a new multiplier

Start with the first block of the LN equation and expand the Lagrangian gradient:

$$\boldsymbol{H}_L^{(k)} \boldsymbol{d}^{(k)} + \boldsymbol{A}^{(k)} \boldsymbol{\delta}^{(k)} = -\nabla f^{(k)} - \boldsymbol{A}^{(k)} \boldsymbol{\lambda}^{(k)}.$$

Move the last term to the left and introduce $\widehat{\boldsymbol{\lambda}}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \boldsymbol{\delta}^{(k)}$. Together with the unchanged constraint equation, this gives

$$\begin{pmatrix} \boldsymbol{H}_L^{(k)} & \boldsymbol{A}^{(k)} \\ (\boldsymbol{A}^{(k)})^{\mathsf{T}} & \boldsymbol{0} \end{pmatrix} \begin{pmatrix} \boldsymbol{d}^{(k)} \\ \widehat{\boldsymbol{\lambda}}^{(k+1)} \end{pmatrix} = - \begin{pmatrix} \nabla f^{(k)} \\ \boldsymbol{g}^{(k)} \end{pmatrix}. \tag{16.10}$$

We next ask whether these equations can arise from a minimization problem in the primal step. The first block is the derivative with respect to $\boldsymbol{d}$ of a quadratic objective, together

with a multiplier term; the second block is an affine constraint. This identifies the model

$$\begin{aligned} \min_{\boldsymbol{d}\in\mathbb{R}^n} \quad & \nabla f(\boldsymbol{x}^{(k)})^{\mathsf{T}}\boldsymbol{d} + \tfrac{1}{2}\boldsymbol{d}^{\mathsf{T}}\boldsymbol{H}_L^{(k)}\boldsymbol{d} \\ \text{subject to} \quad & \boldsymbol{g}^{(k)} + (\boldsymbol{A}^{(k)})^{\mathsf{T}}\boldsymbol{d} = \boldsymbol{0}. \end{aligned} \tag{16.11}$$

Its Lagrangian, with multiplier $\widehat{\boldsymbol{\lambda}}^{(k+1)}$, is

$$\nabla f^{(k)\mathsf{T}}\boldsymbol{d} + \tfrac{1}{2}\boldsymbol{d}^{\mathsf{T}}\boldsymbol{H}_L^{(k)}\boldsymbol{d} + (\widehat{\boldsymbol{\lambda}}^{(k+1)})^{\mathsf{T}}(\boldsymbol{g}^{(k)} + (\boldsymbol{A}^{(k)})^{\mathsf{T}}\boldsymbol{d}).$$

Differentiating with respect to $\boldsymbol{d}$ and imposing the constraint recovers Equation (16.10). A quadratic objective and affine constraints define a quadratic program. Rebuilding this model at successive iterates leads to *sequential quadratic programming* (SQP). We retain QP as the abbreviation for *quadratic penalty* method, and spell out "quadratic program" when referring to an SQP subproblem.

For exact Hessians, the algebraic difference from LN is the multiplier variable:

$$\boldsymbol{\delta}^{(k)} = \widehat{\boldsymbol{\lambda}}^{(k+1)} - \boldsymbol{\lambda}^{(k)}, \qquad \boldsymbol{\lambda}^{(k+1)} = \widehat{\boldsymbol{\lambda}}^{(k+1)} \quad \text{for a full step.} \tag{16.12}$$

If the common matrix is nonsingular, both systems give the same unique primal step. Whenever these stationary steps minimize their quadratic models, exact-Hessian equality SQP and LN generate the same sequence from the same initial pair with full steps. The optimization interpretation does not by itself change that sequence; it supplies a framework for modifying the model and acceptance rule.

16.4.2 Why the quadratic term contains constraint curvature

The linearized constraints have lost the curvature of the original feasible set. To see how the model retains its effect, keep the current multiplier fixed and Taylor-expand $L(\boldsymbol{x}+\boldsymbol{d}, \boldsymbol{\lambda})$. On the linearized feasible set,

$$\nabla_{\boldsymbol{x}} L^{\mathsf{T}}\boldsymbol{d} = \nabla f^{\mathsf{T}}\boldsymbol{d} + \boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{A}^{\mathsf{T}}\boldsymbol{d} = \nabla f^{\mathsf{T}}\boldsymbol{d} - \boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{g}.$$

The last term is constant over the admissible directions. The quadratic term of the same Taylor model is $\boldsymbol{d}^{\mathsf{T}}\boldsymbol{H}_L\boldsymbol{d}/2$, where

$$\boldsymbol{H}_L = \boldsymbol{H}_f + \sum_i \lambda_i \boldsymbol{H}_{g_i}.$$

Removing constants therefore gives precisely the objective in Equation (16.11). The multiplier-weighted constraint Hessians transfer information about the removed constraint curvature into the model objective.

To see this transfer in objective landscapes, consider

$$\min_{\boldsymbol{x}\in\mathbb{R}^2} f(\boldsymbol{x}) = x_2 \qquad \text{subject to} \qquad g(\boldsymbol{x}) = x_1^2 - x_2 = 0.$$

The feasible points have the form (t, t^2), and the objective along this curve is $f(t, t^2) = t^2$. Hence the unique minimizer is $\boldsymbol{x}^* = \boldsymbol{0}$. At this point, $\nabla f = (0, 1)^{\mathsf{T}}$ and $\boldsymbol{A} = (0, -1)^{\mathsf{T}}$, so stationarity gives $\lambda^* = 1$. Although the objective is a plane with $\boldsymbol{H}_f = \boldsymbol{0}$, the Lagrangian Hessian is

$$\boldsymbol{H}_L = \boldsymbol{H}_f + \lambda^* \boldsymbol{H}_g = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

Now construct a quadratic model at $(\boldsymbol{x}^*, \lambda^*)$. The linearized constraint is $d_2 = 0$. If we use only the objective Hessian, the model is $q_f(\boldsymbol{d}) = d_2$, which is zero everywhere on the model's feasible line. It has lost the increasing objective height along the original feasible parabola, and every $(t, 0)$ is a subproblem minimizer. Using the Lagrangian Hessian instead gives

$$q_L(\boldsymbol{d}) = \nabla f^{\mathsf{T}}\boldsymbol{d} + \tfrac{1}{2}\boldsymbol{d}^{\mathsf{T}}\boldsymbol{H}_L\boldsymbol{d} = d_2 + d_1^2, \qquad q_L(t, 0) = t^2.$$

The corrected model now has the unique feasible minimizer $\boldsymbol{d} = \boldsymbol{0}$. Its objective height along the straight feasible line reproduces the height along the curved feasible set of the original problem. The constraint curvature has moved into the objective model rather than disappeared. For a graphical illustration, see Figure 16.2.

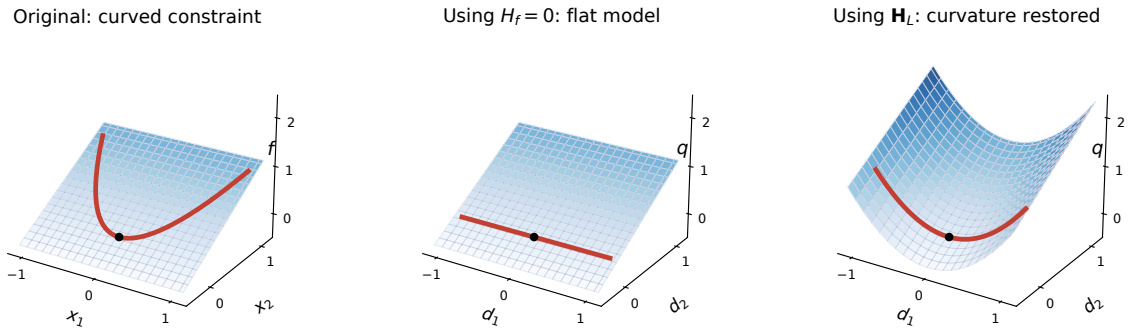


Figure 16.2: Constraint curvature in the original problem and its local models. The vertical coordinate is objective value; the red path shows that value restricted to the feasible curve or line. The original plane gives height t^2 along its curved feasible set. Linearizing the constraint with $\boldsymbol{H}_f$ gives a flat feasible path, whereas the SQP surface built with $\boldsymbol{H}_L$ restores height t^2 along the linearized constraint.

16.4.3 When the stationary subproblem point is a minimizer

The previous algebra identifies KKT equations; we still need to check that the quadratic subproblem has a minimizer. To study this question and allow later curvature modifications, replace $\boldsymbol{H}_L^{(k)}$ by a symmetric matrix $\boldsymbol{B}^{(k)}$. The resulting model has the same linearized constraints and the KKT equations

$$\begin{pmatrix} \boldsymbol{B}^{(k)} & \boldsymbol{A}^{(k)} \\ (\boldsymbol{A}^{(k)})^{\mathsf{T}} & \boldsymbol{0} \end{pmatrix} \begin{pmatrix} \boldsymbol{d}^{(k)} \\ \widehat{\boldsymbol{\lambda}}^{(k+1)} \end{pmatrix} = - \begin{pmatrix} \nabla f^{(k)} \\ \boldsymbol{g}^{(k)} \end{pmatrix}. \tag{16.13}$$

The exact model corresponds to $\boldsymbol{B}^{(k)} = \boldsymbol{H}_L^{(k)}$. To determine when these equations solve the minimization problem, suppress the iteration index and eliminate the linearized constraints.

Lemma 16.2 (A unique minimizer of the quadratic model)**.** *Suppose $\boldsymbol{A}$ has full column rank, $\boldsymbol{B}$ is symmetric, and the columns of $\boldsymbol{Z}$ form a basis of* $\mathrm{Null}(\boldsymbol{A}^{\mathsf{T}})$*. If $\boldsymbol{Z}^{\mathsf{T}}\boldsymbol{B}\boldsymbol{Z} \succ 0$, then*

$$\min_{\boldsymbol{d}} \; q(\boldsymbol{d}) = \nabla f^{\mathsf{T}}\boldsymbol{d} + \tfrac{1}{2}\boldsymbol{d}^{\mathsf{T}}\boldsymbol{B}\boldsymbol{d} \qquad \textit{subject to} \qquad \boldsymbol{A}^{\mathsf{T}}\boldsymbol{d} = -\boldsymbol{g}$$

has a unique global minimizer and a unique multiplier, given by Equation (16.13). For $m = n$, the same conclusion holds without a curvature condition because the feasible step is unique.

Proof. Full column rank makes $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{d} = -\boldsymbol{g}$ consistent for every $\boldsymbol{g}$. Choose one particular solution $\boldsymbol{d}_p$, and let the columns of $\boldsymbol{Z}$ be a basis of $\mathrm{Null}(\boldsymbol{A}^{\mathsf{T}})$. Every feasible step is uniquely $\boldsymbol{d}_p + \boldsymbol{Z}\boldsymbol{y}$. Substitution into the objective yields

$$q(\boldsymbol{d}_p + \boldsymbol{Z}\boldsymbol{y}) = \text{constant} + \boldsymbol{y}^{\mathsf{T}}\boldsymbol{Z}^{\mathsf{T}}(\nabla f + \boldsymbol{B}\boldsymbol{d}_p) + \tfrac{1}{2}\boldsymbol{y}^{\mathsf{T}}(\boldsymbol{Z}^{\mathsf{T}}\boldsymbol{B}\boldsymbol{Z})\boldsymbol{y}.$$

If $\boldsymbol{Z}^{\mathsf{T}}\boldsymbol{B}\boldsymbol{Z}$ is positive definite, this reduced quadratic is strictly convex and coercive, so it has a unique global minimizer. The first-order condition makes $\nabla q(\boldsymbol{d}) = \nabla f + \boldsymbol{B}\boldsymbol{d}$ orthogonal to $\mathrm{Null}(\boldsymbol{A}^{\mathsf{T}})$, giving $\nabla f + \boldsymbol{B}\boldsymbol{d} + \boldsymbol{A}\widehat{\boldsymbol{\lambda}} = \boldsymbol{0}$. Full column rank then makes the multiplier unique as well. When $m = n$, the feasible step is already unique and there is no reduced minimization to perform. □

At a nonlinear KKT pair satisfying LICQ and equality SOSC, the exact reduced Hessian is positive definite, and continuity preserves this property nearby. Away from such a pair, nonsingularity alone does not imply minimization: a nonsingular reduced Hessian with a negative eigenvalue gives a unique stationary point of an unbounded quadratic model. One must therefore distinguish solving the KKT equations from solving the minimization subproblem. Full positive definiteness of $\boldsymbol{B}$ is a convenient stronger safeguard; structured or quasi-Newton approximations can also reduce derivative and factorization costs.

Rank deficiency by itself does not preclude a unique primal minimizer. For example, minimizing $\|\boldsymbol{d}\|^2/2$ subject to the redundant equalities $d_2 = 0$ and $2d_2 = 0$ has the unique solution $\boldsymbol{d} = \boldsymbol{0}$, but nonunique multipliers. Curvature, consistency of the linearized constraints, and multiplier uniqueness must be examined separately.

To turn this local model into a safeguarded iteration, we still need to choose suitable model curvature and decide how far to move along the resulting step. We next introduce a merit function and derive its change along an SQP direction; that calculation will also identify useful curvature safeguards.

16.5 Merit functions and damped SQP

16.5.1 Combining the objective with feasibility

We seek a step-acceptance measure suitable for the original optimization problem. The objective alone ignores feasibility, and the Lagrangian has the saddle geometry examined above. For equality constraints, introduce

$$\phi_\rho(\boldsymbol{x}) = f(\boldsymbol{x}) + \rho\,\|\boldsymbol{g}(\boldsymbol{x})\|_1\,, \qquad \rho > 0. \tag{16.14}$$

This parameter weights trial-step acceptance; the search direction still comes from the constrained quadratic model, rather than minimization of a QP-method or ALM subproblem.

To first see the role of ρ on a scalar problem, minimize $-x$ subject to $x = 0$. Its only feasible point is the minimizer, with $\lambda^* = 1$, and

$$\phi_\rho(x) = -x + \rho|x| = \begin{cases} (\rho - 1)x, & x \geq 0, \\ -(\rho + 1)x, & x < 0. \end{cases}$$

For $\rho > 1$, this function has its unique minimum at zero. For $\rho = 1$, every $x \geq 0$ minimizes it. For $0 < \rho < 1$, it keeps decreasing as x increases away from feasibility. Thus a sufficiently

large weight can make the merit minimum agree with the constrained solution, while a decrease in merit alone need not mean progress toward feasibility.

We first connect merit minimization back to KKT without differentiating the absolute values. Only tangent directions will be needed.

Lemma 16.3 (From feasible merit stationarity to KKT)**.** *Let $f, \boldsymbol{g}$ be continuously differentiable near a feasible point $\boldsymbol{x}$, and let $\rho > 0$. If $D\phi_\rho(\boldsymbol{x}; \boldsymbol{d}) \geq 0$ for every $\boldsymbol{d} \in \mathbb{R}^n$, then $\boldsymbol{x}$ is a KKT point of Equation (16.1). In particular, every feasible local minimizer of ϕ_ρ is a KKT point. No constraint qualification is required.*

Proof. For a tangent direction $\boldsymbol{A}^\mathsf{T}\boldsymbol{d} = \boldsymbol{0}$, the directional derivative reduces to $\nabla f^\mathsf{T}\boldsymbol{d}$. Applying the same condition to $-\boldsymbol{d}$ gives $\nabla f^\mathsf{T}\boldsymbol{d} = 0$. Thus ∇f is orthogonal to $\mathrm{Null}(\boldsymbol{A}^\mathsf{T})$, or equivalently $\nabla f \in \mathrm{Range}(\boldsymbol{A})$. There is therefore a multiplier satisfying $\nabla f + \boldsymbol{A}\boldsymbol{\lambda} = \boldsymbol{0}$, and the feasible point is a KKT point. A local minimum has nonnegative directional derivatives in every direction, so it satisfies the hypothesis as well. □

For the converse, we need enough curvature to control tangent perturbations, as well as a penalty weight that controls departures from feasibility.

Theorem 16.4 (Local exactness under SOSC)**.** *Let $f, \boldsymbol{g}$ be twice continuously differentiable near a feasible KKT pair $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$. If LICQ and the equality SOSC hold at this pair, then every $\rho > \|\boldsymbol{\lambda}^*\|_\infty$ makes $\boldsymbol{x}^*$ a strict local minimizer of ϕ_ρ.*

Proof. First inspect a directional derivative:

$$\begin{aligned} D\phi_\rho(\boldsymbol{x}^*; \boldsymbol{d}) &= \nabla f(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} + \rho \left\| \boldsymbol{A}(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} \right\|_1 \\ &= -\boldsymbol{\lambda}^{*\mathsf{T}} \boldsymbol{A}(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} + \rho \left\| \boldsymbol{A}(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} \right\|_1 \\ &\geq (\rho - \|\boldsymbol{\lambda}^*\|_\infty) \left\| \boldsymbol{A}(\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{d} \right\|_1 . \end{aligned}$$

A first-order move away from the constraint therefore incurs enough penalty to dominate the multiplier-predicted objective gain. Tangent directions give zero in this estimate, so a first-order calculation alone does not finish the proof.

To treat all nearby perturbations together, fix $\boldsymbol{\lambda}^*$ and write

$$\boldsymbol{A}_* = \boldsymbol{A}(\boldsymbol{x}^*), \qquad \boldsymbol{H}_L^* = \nabla^2_{\boldsymbol{x}\boldsymbol{x}} L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*), \qquad \eta = \rho - \|\boldsymbol{\lambda}^*\|_\infty > 0.$$

For a small displacement $\boldsymbol{v}$, expand $L(\cdot, \boldsymbol{\lambda}^*)$ at $\boldsymbol{x}^*$. The KKT equalities

$$\nabla_x L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \boldsymbol{0}, \qquad \boldsymbol{g}(\boldsymbol{x}^*) = \boldsymbol{0}$$

and Hölder's inequality give

$$\begin{aligned} \phi_\rho(\boldsymbol{x}^* + \boldsymbol{v}) - \phi_\rho(\boldsymbol{x}^*) &= L(\boldsymbol{x}^* + \boldsymbol{v}, \boldsymbol{\lambda}^*) - L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) \\ &\quad - \boldsymbol{\lambda}^{*\mathsf{T}} \boldsymbol{g}(\boldsymbol{x}^* + \boldsymbol{v}) + \rho \|\boldsymbol{g}(\boldsymbol{x}^* + \boldsymbol{v})\|_1 \\ &\geq \tfrac{1}{2}\boldsymbol{v}^\mathsf{T}\boldsymbol{H}_L^*\boldsymbol{v} + o(\|\boldsymbol{v}\|_2^2) + \eta \|\boldsymbol{g}(\boldsymbol{x}^* + \boldsymbol{v})\|_1 . \end{aligned} \tag{16.15}$$

The Hessian $\boldsymbol{H}_L^*$ need not be positive definite on the whole space, so the sign of this lower bound still requires an argument.

Suppose, to obtain a contradiction, that strict local minimality fails. There are then nonzero displacements $\boldsymbol{v}_j \to \mathbf{0}$ such that $\phi_\rho(\boldsymbol{x}^* + \boldsymbol{v}_j) \le \phi_\rho(\boldsymbol{x}^*)$. The quadratic term and the Taylor remainder have absolute value bounded by $C \|\boldsymbol{v}_j\|_2^2$ for some fixed $C > 0$ and all sufficiently large j. Hence (16.15) first implies

$$\eta \|\boldsymbol{g}(\boldsymbol{x}^* + \boldsymbol{v}_j)\|_1 \le C \|\boldsymbol{v}_j\|_2^2 .$$

Normalize these displacements by setting $\boldsymbol{d}_j = \boldsymbol{v}_j / \|\boldsymbol{v}_j\|_2$. Compactness of the unit sphere gives a subsequence, still indexed by j, with $\boldsymbol{d}_j \to \boldsymbol{d}$ and $\|\boldsymbol{d}\|_2 = 1$. The constraint expansion

$$\boldsymbol{g}(\boldsymbol{x}^* + \boldsymbol{v}_j) = \boldsymbol{A}_*^\mathsf{T} \boldsymbol{v}_j + O(\|\boldsymbol{v}_j\|_2^2)$$

shows, after division by $\|\boldsymbol{v}_j\|_2$, that $\boldsymbol{A}_*^\mathsf{T} \boldsymbol{d}_j = O(\|\boldsymbol{v}_j\|_2) \to \mathbf{0}$. Thus $\boldsymbol{A}_*^\mathsf{T} \boldsymbol{d} = \mathbf{0}$: the normalized nonascending displacements converge to a nonzero tangent direction.

Finally, divide (16.15) by $\|\boldsymbol{v}_j\|_2^2$, discard its nonnegative penalty term, and pass to the limit. We obtain

$$0 \ge \tfrac{1}{2} \boldsymbol{d}^\mathsf{T} \boldsymbol{H}_L^* \boldsymbol{d} > 0,$$

where the strict inequality is SOSC. This contradiction proves the claimed strict local merit minimum. The penalty margin $\eta > 0$ forces any potentially nonascending sequence toward the tangent space; SOSC then rules out its limiting direction. □

This is the local *exact penalty* property: a finite penalty weight can already make the constrained solution a merit minimizer. In contrast, a squared penalty contributes no first-order change at a feasible point. The local coordinate argument in Theorem 16.6 gives a complementary, more general result: LICQ and constrained local minimality suffice for local exactness, without requiring SOSC; strict constrained minimality gives strict merit minimality. Once a KKT multiplier and SOSC are given, the Taylor proof above does not otherwise use LICQ.

These implications concern local optimality, not just decrease of merit values. A decreasing sequence alone guarantees neither feasibility nor merit stationarity. A convergence argument must additionally control the computed direction, feasibility, step lengths, and multiplier estimates. The descent calculation below makes these requirements explicit.

16.5.2 Deriving the small-step change

We next examine the merit change along a step that satisfies the linearized constraints. This will give a condition that can be checked after solving the quadratic model.

Lemma 16.5 (Merit change along an SQP step)**.** *Let $f, \boldsymbol{g}$ be twice continuously differentiable near $\boldsymbol{x}$, fix $\rho > 0$, and suppose $\boldsymbol{A}^\mathsf{T} \boldsymbol{d} = -\boldsymbol{g}$. Then*

$$D\phi_\rho(\boldsymbol{x}; \boldsymbol{d}) = \nabla f^\mathsf{T} \boldsymbol{d} - \rho \|\boldsymbol{g}\|_1 .$$

If a symmetric $\boldsymbol{B}$ and a multiplier $\widehat{\boldsymbol{\lambda}}$ also satisfy $\nabla f + \boldsymbol{B}\boldsymbol{d} + \boldsymbol{A}\widehat{\boldsymbol{\lambda}} = \mathbf{0}$, then

$$D\phi_\rho(\boldsymbol{x}; \boldsymbol{d}) \le -\boldsymbol{d}^\mathsf{T} \boldsymbol{B} \boldsymbol{d} - (\rho - \|\widehat{\boldsymbol{\lambda}}\|_\infty) \|\boldsymbol{g}\|_1 .$$

In particular, $\boldsymbol{B} \succ 0$, $\boldsymbol{d} \ne \mathbf{0}$, and $\rho > \|\widehat{\boldsymbol{\lambda}}\|_\infty$ guarantee strict merit descent.

Proof. Taylor's theorem yields, for $0 \le \alpha \le 1$ sufficiently small,

$$f(\boldsymbol{x} + \alpha \boldsymbol{d}) - f(\boldsymbol{x}) = \alpha \nabla f(\boldsymbol{x})^{\mathsf{T}} \boldsymbol{d} + O(\alpha^2 \left\| \boldsymbol{d} \right\|^2),$$

and Equation (16.7) gives $\boldsymbol{g}(\boldsymbol{x} + \alpha \boldsymbol{d}) = (1 - \alpha)\boldsymbol{g} + \boldsymbol{r}(\alpha)$, with $\left\| \boldsymbol{r}(\alpha) \right\|_1 = O(\alpha^2 \left\| \boldsymbol{d} \right\|^2)$. Use the reverse triangle inequality to estimate the norm without differentiating absolute values:

$$\left| \left\| (1 - \alpha)\boldsymbol{g} + \boldsymbol{r}(\alpha) \right\|_1 - (1 - \alpha) \left\| \boldsymbol{g} \right\|_1 \right| \le \left\| \boldsymbol{r}(\alpha) \right\|_1 .$$

Combining these estimates gives

$$\phi_\rho(\boldsymbol{x} + \alpha \boldsymbol{d}) - \phi_\rho(\boldsymbol{x}) = \alpha(\nabla f^{\mathsf{T}} \boldsymbol{d} - \rho \left\| \boldsymbol{g} \right\|_1) + O(\alpha^2 \left\| \boldsymbol{d} \right\|^2). \tag{16.16}$$

The remainder constant may depend on the fixed ρ and local derivative bounds. Dividing by $\alpha > 0$ and taking a limit shows that

$$D\phi_\rho(\boldsymbol{x}; \boldsymbol{d}) = \nabla f^{\mathsf{T}} \boldsymbol{d} - \rho \left\| \boldsymbol{g} \right\|_1 .$$

In particular, at a feasible point the constraint violation contributes only a second-order term. If $\nabla f^{\mathsf{T}} \boldsymbol{d} < 0$, the negative first-order objective change dominates for sufficiently small positive α, whatever finite value of ρ is fixed.

To obtain a useful sufficient condition at an infeasible point, use subproblem stationarity $\nabla f + \boldsymbol{B}\boldsymbol{d} + \boldsymbol{A}\widehat{\boldsymbol{\lambda}} = \boldsymbol{0}$:

$$\begin{aligned} \nabla f^{\mathsf{T}} \boldsymbol{d} &= -\boldsymbol{d}^{\mathsf{T}} \boldsymbol{B} \boldsymbol{d} - \widehat{\boldsymbol{\lambda}}^{\mathsf{T}} \boldsymbol{A}^{\mathsf{T}} \boldsymbol{d} = -\boldsymbol{d}^{\mathsf{T}} \boldsymbol{B} \boldsymbol{d} + \widehat{\boldsymbol{\lambda}}^{\mathsf{T}} \boldsymbol{g}, \\ D\phi_\rho(\boldsymbol{x}; \boldsymbol{d}) &\le -\boldsymbol{d}^{\mathsf{T}} \boldsymbol{B} \boldsymbol{d} - (\rho - \left\| \widehat{\boldsymbol{\lambda}} \right\|_\infty) \left\| \boldsymbol{g} \right\|_1 . \end{aligned}$$

Under the stated positive-definiteness and parameter conditions, the first term is strictly negative and the second is nonpositive, giving strict descent. □

More generally, the displayed bound suffices whenever its right-hand side is negative. Under the same condition $\rho > \left\| \widehat{\boldsymbol{\lambda}} \right\|_\infty$, positive semidefiniteness suffices if either $\boldsymbol{d}^{\mathsf{T}} \boldsymbol{B} \boldsymbol{d} > 0$ or $\boldsymbol{g} \ne \boldsymbol{0}$. Positive curvature only on tangent directions does not by itself make this particular bound negative at an infeasible point, because the step need not be tangent. A robust implementation checks the actual predicted derivative or uses an appropriate curvature safeguard.

The Taylor argument uses only homogeneity and the reverse triangle inequality, so it also holds for $f + \rho \left\| \boldsymbol{g} \right\|$ with any norm, including the Euclidean norm. The multiplier estimate then uses the corresponding dual norm: $\widehat{\boldsymbol{\lambda}}^{\mathsf{T}} \boldsymbol{g} \le \left\| \widehat{\boldsymbol{\lambda}} \right\|_* \left\| \boldsymbol{g} \right\|$. For $\left\| \boldsymbol{g} \right\|_2$, the sufficient parameter bound becomes $\rho > \left\| \widehat{\boldsymbol{\lambda}} \right\|_2$. For the common test below, the single constraint makes the one-norm and two-norm penalties identical.

16.5.3 Regularizing the quadratic model

The merit-descent bound shows how model curvature affects step acceptance. We now use the optimization interpretation to enforce a suitable curvature condition before computing the SQP step. Consider the shifted matrix

$$\boldsymbol{B} = \boldsymbol{H}_L + \tau \boldsymbol{I}, \qquad \tau \ge 0.$$

When the tangent space is nonzero, choose an orthonormal basis $\boldsymbol{Z}$. The reduced Hessian is $\boldsymbol{Z}^{\mathsf{T}} \boldsymbol{H}_L \boldsymbol{Z} + \tau \boldsymbol{I}$. A shift greater than the negative of the smallest reduced eigenvalue makes

this matrix positive definite. A larger shift can make $\boldsymbol{B}$ positive definite on the whole primal space. This stronger condition makes $-\boldsymbol{d}^{\mathsf{T}}\boldsymbol{B}\boldsymbol{d} < 0$ for every nonzero step, as used in the preceding descent bound, including at an infeasible point. Positive reduced curvature alone guarantees model minimization but need not make that bound negative. If the original curvature is already suitable, no shift is needed.

The shift acts on the quadratic objective in $\boldsymbol{d}$ and leaves $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{d} = -\boldsymbol{g}$ unchanged. This differs from shifting the entire KKT matrix: adding a multiplier–multiplier diagonal block would change the linearized constraint equation itself. A suitable Hessian approximation is another way to obtain a well-posed model when exact second derivatives are expensive. Further merit and feasibility safeguards are discussed in Section 16.8.

After choosing $\boldsymbol{B}$, solve the quadratic model for the step and its multiplier. We can then check the merit parameter in the descent bound before choosing a step length. This separates modifying the model from selecting the acceptance rule for the original nonlinear problem.

16.5.4 Checking the merit parameter after solving the model

The descent bound contains $\widehat{\boldsymbol{\lambda}}^{(k+1)}$, which is not known until the current quadratic subproblem has been solved. To see how to use the bound, inspect Equation (16.13): its coefficients depend on the current pair and the chosen $\boldsymbol{B}^{(k)}$, but do not contain ρ. We can therefore first compute the direction and subproblem multiplier, then choose or check the merit parameter, and finally run a line search. Changing ρ at this stage changes the acceptance function without requiring another solution of the same quadratic model.

The two multiplier bounds in this section serve different purposes. The $\boldsymbol{\lambda}^*$ in the exactness argument belongs to a solution of the original nonlinear problem. The $\widehat{\boldsymbol{\lambda}}^{(k+1)}$ in the descent argument belongs to the current quadratic model and becomes available at every iteration. With $\boldsymbol{B}^{(k)} \succ 0$ and $\boldsymbol{d}^{(k)} \neq \boldsymbol{0}$, choosing $\rho > \left\|\widehat{\boldsymbol{\lambda}}^{(k+1)}\right\|_\infty$ guarantees strict descent for that iteration. The multiplier bound alone does not give this conclusion without the required curvature condition.

For the basic algorithm we instead fix ρ at initialization. Once the direction has been computed, evaluate the actual directional derivative

$$D^{(k)} = (\nabla f^{(k)})^{\mathsf{T}}\boldsymbol{d}^{(k)} - \rho\left\|\boldsymbol{g}^{(k)}\right\|_1 .$$

This calculation does not require a multiplier estimate, although the model solve has already supplied one. If $D^{(k)} < 0$, the direction is suitable for a descent line search. The sufficient bound on $\left\|\widehat{\boldsymbol{\lambda}}^{(k+1)}\right\|_\infty$ need not hold: its failure does not imply that $D^{(k)} \geq 0$. At a feasible point, for example, $D^{(k)} = -(\boldsymbol{d}^{(k)})^{\mathsf{T}}\boldsymbol{B}^{(k)}\boldsymbol{d}^{(k)}$, so positive curvature on the nonzero tangent step gives descent for every $\rho > 0$.

If $D^{(k)} \geq 0$ and the KKT test has not passed, the basic implementation reports a non-descent direction. Backtracking only shortens the step; it does not change this directional derivative or supply the missing descent guarantee. An adaptive merit-parameter rule is a possible extension, described in Section 16.8.3.

16.5.5 Backtracking and the distinction between small and full steps

If $D = D\phi_\rho(\boldsymbol{x};\boldsymbol{d}) < 0$, choose $0 < \sigma < 1$ and accept a trial step satisfying the Armijo condition

$$\phi_\rho(\boldsymbol{x} + \alpha\boldsymbol{d}) \le \phi_\rho(\boldsymbol{x}) + \sigma\alpha D. \tag{16.17}$$

Indeed, Equation (16.16) bounds the difference on the left by $\alpha D + C\alpha^2 \|\boldsymbol{d}\|^2$. For nonzero $\boldsymbol{d}$, it is enough to take

$$0 < \alpha \le \frac{(1-\sigma)(-D)}{C\|\boldsymbol{d}\|^2},$$

within the neighbourhood where the Taylor bound is valid. Therefore backtracking from 1 by a fixed factor $\beta \in (0,1)$ eventually succeeds, provided trial points stay in the domain. This proves small-step acceptance, not acceptance of $\alpha = 1$, and not global convergence of the entire algorithm.

We can see the difference exactly on the first step of Equation (16.8). With $\rho = 2$ and $\boldsymbol{d} = (-3/8, 3/8)^{\mathsf{T}}$,

$$\nabla f(\boldsymbol{x}^{(0)})^{\mathsf{T}}\boldsymbol{d} = -9/32, \qquad g(\boldsymbol{x}^{(0)} + \alpha\boldsymbol{d}) = 9\alpha^2/64,$$

so direct expansion gives

$$\phi_2(\boldsymbol{x}^{(0)} + \alpha\boldsymbol{d}) - \phi_2(\boldsymbol{x}^{(0)}) = -\frac{9}{32}\alpha + \frac{27}{64}\alpha^2. \tag{16.18}$$

The full step increases the merit value by $9/64$, although its directional derivative is negative. Strict decrease holds precisely for $0 < \alpha < 2/3$. The Armijo condition holds for $0 < \alpha \le 2(1-\sigma)/3$; for example, with $\sigma = 10^{-4}$, halving rejects 1 and accepts $1/2$. This calculation displays explicitly how first-order progress can be outweighed by second-order constraint error when the step is too large.

16.5.6 The damped SQP algorithm

After choosing an acceptable α, update the primal point and interpolate toward the subproblem multiplier:

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \alpha\boldsymbol{d}^{(k)}, \qquad \boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \alpha(\widehat{\boldsymbol{\lambda}}^{(k+1)} - \boldsymbol{\lambda}^{(k)}). \tag{16.19}$$

The full step takes the subproblem multiplier directly. For a damped step, $\widehat{\boldsymbol{\lambda}}^{(k+1)}$ and the accepted $\boldsymbol{\lambda}^{(k+1)}$ are different quantities. Interpolating both components also recovers a damped Newton step when $\boldsymbol{B} = \boldsymbol{H}_L$.

Algorithm: Basic damped equality SQP

Choose $(\boldsymbol{x}^{(0)}, \boldsymbol{\lambda}^{(0)})$, a merit parameter $\rho > 0$, $\sigma, \beta \in (0,1)$, and $\varepsilon_{\text{out}} > 0$. At each outer iteration:

1. Compute r_{KKT} from Equation (16.4) and stop if it is at most ε_{out}.
2. Form a symmetric $\boldsymbol{B}^{(k)}$ and the linearized constraints. Check that the quadratic model is well posed, and solve Equation (16.13) for $(\boldsymbol{d}^{(k)}, \widehat{\boldsymbol{\lambda}}^{(k+1)})$.
3. If $(\boldsymbol{x}^{(k)}, \widehat{\boldsymbol{\lambda}}^{(k+1)})$ already meets the original KKT tolerance, retain $\boldsymbol{x}^{(k)}$, accept that multiplier, and stop. Otherwise compute $D = \nabla f^{(k)\mathsf{T}}\boldsymbol{d}^{(k)} - \rho\left\|\boldsymbol{g}^{(k)}\right\|_1$ and

require $D < 0$.

4. Starting at $\alpha = 1$, multiply α by β until the trial point is in the domain and satisfies Equation (16.17).
5. Take Equation (16.19) and recompute the original KKT residual.

If the curvature check or the condition $D < 0$ fails, report the failure and stop this basic scheme. Likewise, report a singular system, failed line search, or exhausted update budget. Check the final returned pair against the original KKT tolerance.

The function `damped_sqp` implements this scheme in `l16_lagrange_newton_sqp.py`; its `equality_sqp_step` function solves for the direction and the new subproblem multiplier. The default uses the exact Hessian, fixed $\rho = 2$, $\sigma = 10^{-4}$, and $\beta = 1/2$. An optional fixed shift specifies $\boldsymbol{B} = \boldsymbol{H}_L + \tau \boldsymbol{I}$; its default is $\tau = 0$. The code checks reduced curvature and predicted merit descent and reports failure if either check fails. It does not automatically change the shift or merit parameter.

16.5.7 Why a line search can interfere with the local rate

Having specified the damped iteration, let us check whether it inherits the local quadratic rate of LN. The Newton proof uses full steps, so this requires examining whether the line search eventually accepts $\alpha = 1$. Consider

$$\min_{\boldsymbol{x} \in \mathbb{R}^2} 2(x_1^2 + x_2^2 - 1) - x_1 \qquad \text{subject to} \qquad x_1^2 + x_2^2 - 1 = 0.$$

On the circle the objective is $-x_1$, giving $\boldsymbol{x}^* = (1, 0)^\mathsf{T}$. The derivatives are

$$\nabla f = 4\boldsymbol{x} - (1, 0)^\mathsf{T}, \qquad \boldsymbol{A} = 2\boldsymbol{x}, \qquad \boldsymbol{H}_L = (4 + 2\lambda)\boldsymbol{I}.$$

Stationarity at $\boldsymbol{x}^*$ gives $\lambda^* = -3/2$ and hence $\boldsymbol{H}_L = \boldsymbol{I}$; LICQ and SOSC hold. For each small nonzero θ, take a separate feasible starting pair

$$\boldsymbol{x}_\theta = (\cos\theta, \sin\theta)^\mathsf{T}, \qquad \lambda = \lambda^*.$$

The linearized constraint requires $\boldsymbol{x}_\theta^\mathsf{T} \boldsymbol{d} = 0$. Projecting the subproblem stationarity equation

$$\boldsymbol{d} + 4\boldsymbol{x}_\theta - (1, 0)^\mathsf{T} + 2\boldsymbol{x}_\theta \widehat{\lambda} = \boldsymbol{0}$$

onto the tangent and normal spaces gives, respectively,

$$\boldsymbol{d}_\theta = (\sin^2\theta, -\sin\theta\cos\theta)^\mathsf{T}, \qquad \widehat{\lambda}_\theta = (\cos\theta - 4)/2.$$

In particular, $\|\boldsymbol{d}_\theta\|^2 = \sin^2\theta$. Substitution along the trial line, for $\alpha \geq 0$, now yields

$$\begin{aligned} g(\boldsymbol{x}_\theta + \alpha \boldsymbol{d}_\theta) &= \alpha^2 \sin^2\theta, \\ f(\boldsymbol{x}_\theta + \alpha \boldsymbol{d}_\theta) - f(\boldsymbol{x}_\theta) &= (-\alpha + 2\alpha^2)\sin^2\theta, \\ \phi_\rho(\boldsymbol{x}_\theta + \alpha \boldsymbol{d}_\theta) - \phi_\rho(\boldsymbol{x}_\theta) &= [-\alpha + (2 + \rho)\alpha^2]\sin^2\theta. \end{aligned}$$

Thus $D = -\sin^2\theta < 0$, but the full step raises the merit value by $(1 + \rho)\sin^2\theta > 0$. To assess the quality of that rejected step, compute its distance to the solution:

$$\|\boldsymbol{x}_\theta + \boldsymbol{d}_\theta - \boldsymbol{x}^*\| = 1 - \cos\theta = \tfrac{1}{2}\|\boldsymbol{x}_\theta - \boldsymbol{x}^*\|^2.$$

It would reduce the primal error quadratically. Nevertheless, Armijo requires

$$\alpha \leq \frac{1-\sigma}{2+\rho},$$

whose upper bound does not approach one as $\theta \to 0$. For $\rho = 2$, $\sigma = 10^{-4}$, and halving, the accepted step is $1/8$ for every member of this family. For a fixed $\alpha < 1$, the new second coordinate is $\sin\theta(1 - \alpha\cos\theta)$, which remains first order in θ. See Figure 16.3. The damped step therefore loses the quadratic primal-error reduction of the full step.

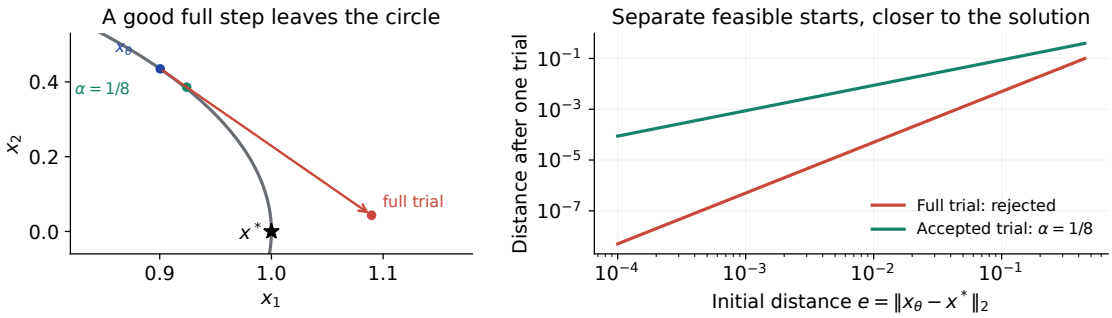


Figure 16.3: The Maratos obstruction at feasible starting points approaching the solution. Each point is a separate one-step probe, not part of one SQP iteration history. Full steps reduce the primal error quadratically but fail the merit test.

This obstruction is the *Maratos effect*: second-order constraint error can outweigh the objective improvement in a merit test, even arbitrarily near a regular solution. The first-step rejection in our common test demonstrates the same competing terms, but a single rejection away from the solution would not by itself establish this local obstruction.

The family proves that arbitrarily close starting pairs can reject full steps; it does not assert that one damped SQP trajectory stays on this family or rejects every subsequent full step. A second-order correction can reduce the remaining constraint error while preserving the leading Newton step. Corrections or alternative acceptance mechanisms, with their own analysis, can restore eventual full-step acceptance.

16.6 Comparing pure LN and damped SQP on the common test

Use Equation (16.8) with the same initial primal–multiplier pair and $\varepsilon_{\text{out}} = 10^{-6}$. Both methods use direct linear solves and recompute the nonlinear residual after every accepted update. For pure LN the first step is accepted in full; for damped SQP, the calculation in Equation (16.18) makes the first accepted step length $1/2$. After that first decision the methods evaluate their Hessians and constraints at different points, so their later directions need not coincide.

The function `run_comparison` runs both methods and prints their residuals, KKT solve counts, and accepted step lengths. To evaluate the Armijo test accurately near the solution, write $\boldsymbol{s} = \alpha\boldsymbol{d}$ and use the exact identities for this common problem:

$$f(\boldsymbol{x}+\boldsymbol{s}) - f(\boldsymbol{x}) = \nabla f(\boldsymbol{x})^{\mathsf{T}}\boldsymbol{s} + \tfrac{1}{2}\|\boldsymbol{s}\|^2, \qquad g(\boldsymbol{x}+\boldsymbol{s}) = g(\boldsymbol{x}) + \boldsymbol{A}(\boldsymbol{x})^{\mathsf{T}}\boldsymbol{s} + s_1^2.$$

Substituting these into the merit difference avoids subtracting nearly equal objective values. No Taylor remainder is discarded in this evaluation. The implementation contains the two algorithms and this comparison experiment; the figures below display their recorded histories.

Method	KKT solves	Accepted step lengths	r_{stat}	r_{feas}
Pure LN	4	$1, 1, 1, 1$	7.322×10^{-7}	8.160×10^{-8}
Damped SQP	3	$1/2, 1, 1$	7.082×10^{-8}	5.670×10^{-8}

For reference, the first accepted SQP pair is $\boldsymbol{x}^{(1)} = (0.3125, 0.9375)^{\mathsf{T}}$, $\lambda^{(1)} = 0.4375$. Its next two updates reach $\boldsymbol{x}^{(3)} \approx (0.312908439, 0.902088365)^{\mathsf{T}}$, $\lambda^{(3)} \approx 1.097911635$. In this run, damping the first step saves one later KKT solve. The line search also requires trial objective and constraint evaluations, which are not counted as KKT solves.

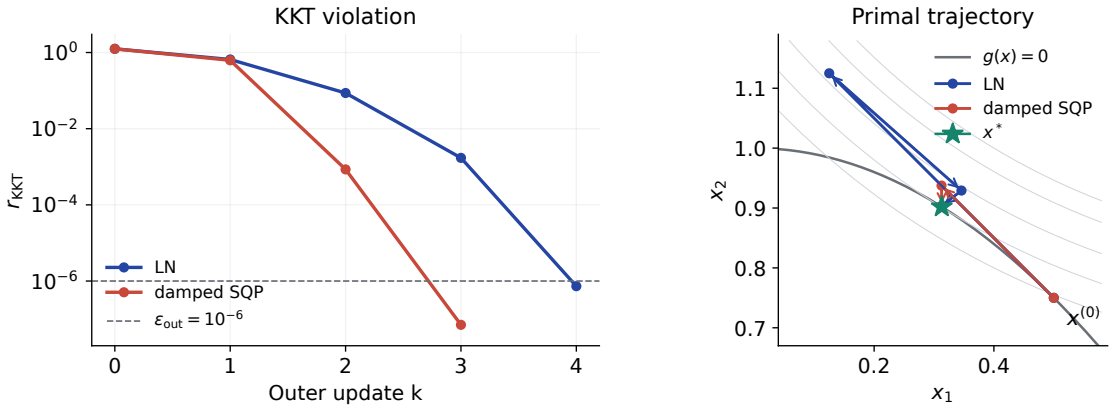


Figure 16.4: Pure LN and basic damped SQP on the same problem and initial pair. The left panel compares original KKT violations; the right panel compares accepted primal iterates. The merit function controls step acceptance, while the reported stopping criterion remains the nonlinear KKT violation.

These counts measure saddle-point solves, not unconstrained inner minimizations as in the QP/ALM experiments of Chapter 15. The comparison illustrates the cost of controlling a full Newton step on one problem. It does not establish a universal ranking of the methods or a global convergence guarantee for these fixed-parameter teaching implementations.

16.7 Implementation and the inequality extension

Before extending the model, it is useful to separate the work required by the methods already studied. QP and ALM repeatedly minimize nonlinear unconstrained subproblems. LN/SQP instead form one local model and solve its saddle-point system. Their model systems contain no diverging quadratic penalty, but depend more directly on local constraint geometry. Compare derivative evaluations, inner work, linear solve sizes, and line-search trials as well as outer counts. A good local model can make LN/SQP very fast near a regular solution; a poor initial point can require globalization or model changes. SQP is consequently one of the standard general frameworks for smooth constrained nonlinear optimization, rather than one single choice of Hessian, linear solver, and acceptance rule.

For damped SQP, merit-parameter selection and line-search work must also be accounted for. Unlike the QP penalty, the merit parameter need not tend to infinity along a regular

convergent run. Pure LN has no penalty parameter. In addition to outer accuracy, inner error, and the acceptance parameter where applicable, LN/SQP require attention to the well-posedness and numerical solution of the KKT system.

In an implementation, solve the KKT system without forming its inverse, monitor Jacobian rank and conditioning, and scale variables and constraints sensibly. For an iterative subproblem solver, $\varepsilon_{\text{in}}^{(k)}$ bounds the subproblem's residual and is distinct from ε_{out}, which bounds the original nonlinear KKT residual. The small dense examples use direct linear solves. For a system $\boldsymbol{K}\boldsymbol{z} = \boldsymbol{b}$, the default implementation checks the normalized linear residual

$$\frac{\|\boldsymbol{K}\boldsymbol{z} - \boldsymbol{b}\|_\infty}{1 + \|\boldsymbol{b}\|_\infty} \le \varepsilon_{\text{in}}^{(k)} \equiv 10^{-10}.$$

This is an inner accuracy check, separate from the original nonlinear KKT residual. Both drivers accept `inner_tol`, and the command-line option `-inner-tol` selects its fixed value. The code does not run an extra inner optimization loop for LN/SQP. The descent identities above assume an exact model solve; with inexact solves, the linearized feasibility and stationarity errors enter those identities and must be controlled. A small linear residual alone is also not a conditioning guarantee for the KKT matrix. Failure of a direct solve or an unexpectedly small step should produce a diagnostic rather than a false success report.

16.7.1 The inequality-constrained model

For mixed constraints $\boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}$, $\boldsymbol{h}(\boldsymbol{x}) \le \boldsymbol{0}$, let $\boldsymbol{C} = [\nabla h_1, \dots, \nabla h_p]$ and use $L = f + \boldsymbol{\lambda}^\mathsf{T}\boldsymbol{g} + \boldsymbol{\nu}^\mathsf{T}\boldsymbol{h}$. Linearizing each constraint gives the SQP model

$$\begin{aligned} \min_{\boldsymbol{d}} \quad & \nabla f^\mathsf{T}\boldsymbol{d} + \tfrac{1}{2}\boldsymbol{d}^\mathsf{T}\boldsymbol{B}\boldsymbol{d} \\ \text{subject to} \quad & \boldsymbol{g} + \boldsymbol{A}^\mathsf{T}\boldsymbol{d} = \boldsymbol{0}, \\ & \boldsymbol{h} + \boldsymbol{C}^\mathsf{T}\boldsymbol{d} \le \boldsymbol{0}. \end{aligned}$$

The subproblem multipliers satisfy

$$\begin{aligned} \nabla f + \boldsymbol{B}\boldsymbol{d} + \boldsymbol{A}\widehat{\boldsymbol{\lambda}} + \boldsymbol{C}\widehat{\boldsymbol{\nu}} &= \boldsymbol{0}, \\ \widehat{\boldsymbol{\nu}} &\ge \boldsymbol{0}, \\ \widehat{\nu}_j(h_j + \nabla h_j^\mathsf{T}\boldsymbol{d}) &= 0. \end{aligned}$$

Together with linearized feasibility, these are the KKT equations of the quadratic model. They do not assert feasibility or complementarity for the original nonlinear constraints at the updated point. Complementarity is the main additional difficulty: neither the binding constraints nor the zero multipliers are known in advance, so the equations no longer reduce to one fixed equality saddle-point system. The active inequalities of the subproblem predict a nonlinear active set, and its multipliers supply new estimates. An active-set solver maintains a working set, adjusts it when a blocking inequality is met or a multiplier has the wrong sign, and solves equality-constrained quadratic subproblems within that process.

To stop a mixed-constraint method, use all four components from Chapter 15:

$$\begin{aligned} r_{\text{feas}} &= \max\{\|\boldsymbol{g}\|_\infty, \|[\boldsymbol{h}]_+\|_\infty\}, & r_{\text{stat}} &= \|\nabla f + \boldsymbol{A}\boldsymbol{\lambda} + \boldsymbol{C}\boldsymbol{\nu}\|_\infty, \\ r_{\text{dual}} &= \|[-\boldsymbol{\nu}]_+\|_\infty, & r_{\text{comp}} &= \|\boldsymbol{\nu} \odot \boldsymbol{h}\|_\infty. \end{aligned}$$

The maximum must be at most ε_{out}. These checks target first-order KKT conditions and are distinct from verifying a second-order optimality certificate.

16.8 Supplementary details and safeguards

The preceding sections specify the basic methods and their numerical comparison. We now retain several further arguments that clarify the assumptions or motivate extensions.

16.8.1 A local coordinate proof of exactness

We now obtain local exactness directly from constrained local minimality. This argument replaces the SOSC assumption in Theorem 16.4 by a local coordinate description of the feasible set.

Theorem 16.6 (Local exactness under LICQ)**.** *Let $f, \boldsymbol{g}$ be continuously differentiable near a constrained local minimizer $\boldsymbol{x}^*$ satisfying LICQ, and let $\boldsymbol{\lambda}^*$ be its KKT multiplier. For every $\rho > \|\boldsymbol{\lambda}^*\|_\infty$, the point $\boldsymbol{x}^*$ locally minimizes ϕ_ρ. If the constrained local minimum is strict, this merit minimum is also strict.*

Proof. By the inverse function theorem, complete the local constraint coordinates $\boldsymbol{v} = \boldsymbol{g}(\boldsymbol{x})$ with tangent coordinates $\boldsymbol{u}$, obtaining a continuously differentiable local parametrization $\boldsymbol{x} = X(\boldsymbol{u}, \boldsymbol{v})$, where $X(\boldsymbol{u}^*, \boldsymbol{0}) = \boldsymbol{x}^*$ and $\boldsymbol{g}(X(\boldsymbol{u}, \boldsymbol{v})) = \boldsymbol{v}$. Set $\psi(\boldsymbol{u}, \boldsymbol{v}) = f(X(\boldsymbol{u}, \boldsymbol{v}))$. Differentiating the last identity with respect to $\boldsymbol{v}$ gives $\boldsymbol{A}(\boldsymbol{x}^*)^\mathsf{T} X_{\boldsymbol{v}} = \boldsymbol{I}$. Using $\nabla f(\boldsymbol{x}^*) = -\boldsymbol{A}(\boldsymbol{x}^*)\boldsymbol{\lambda}^*$, we obtain

$$\nabla_{\boldsymbol{v}} \psi(\boldsymbol{u}^*, \boldsymbol{0}) = X_{\boldsymbol{v}}^\mathsf{T} \nabla f(\boldsymbol{x}^*) = -\boldsymbol{\lambda}^*.$$

Fix $\rho > \|\boldsymbol{\lambda}^*\|_\infty$, and choose $c \in (\|\boldsymbol{\lambda}^*\|_\infty, \rho)$. Continuity gives a small coordinate box in which $\|\nabla_{\boldsymbol{v}} \psi\|_\infty \le c$. Integrating in the normal coordinate yields

$$\psi(\boldsymbol{u}, \boldsymbol{v}) - \psi(\boldsymbol{u}, \boldsymbol{0}) = \int_0^1 \nabla_{\boldsymbol{v}} \psi(\boldsymbol{u}, t\boldsymbol{v})^\mathsf{T} \boldsymbol{v} \, dt \ge -c \|\boldsymbol{v}\|_1 .$$

Since $X(\boldsymbol{u}, \boldsymbol{0})$ is feasible and nearby, local minimality gives $\psi(\boldsymbol{u}, \boldsymbol{0}) \ge f(\boldsymbol{x}^*)$. Therefore

$$\phi_\rho(X(\boldsymbol{u}, \boldsymbol{v})) \ge f(\boldsymbol{x}^*) + (\rho - c) \|\boldsymbol{v}\|_1 \ge f(\boldsymbol{x}^*).$$

This proves local exactness. If the constrained local minimum is strict, then for a nearby point different from $\boldsymbol{x}^*$, either $\boldsymbol{v} \neq \boldsymbol{0}$, which makes the penalty-margin term strictly positive, or $\boldsymbol{v} = \boldsymbol{0}$, which gives strict inequality by constrained minimality. Thus the merit minimum is also strict in a sufficiently small neighbourhood. □

The proof uses local constrained minimality and LICQ; SOSC is one earlier sufficient condition for that minimality. It is not a claim that every local merit minimum is feasible for a parameter selected without reference to the problem.

16.8.2 What a merit convergence argument additionally needs

To see how the one-step descent bound can yield a limiting KKT condition, we make the required uniform assumptions explicit. This is a sufficient framework for the argument, rather than a guarantee for every SQP implementation.

Theorem 16.7 (KKT accumulation points under uniform safeguards)**.** *Consider an infinite SQP sequence with exact model equations Equation (16.13), a fixed $\rho > 0$, and*

Armijo backtracking from 1 *with fixed* $\beta, \sigma \in (0,1)$. *Suppose that:*

- *the primal iterates lie in a compact subset of* $\mathcal{D}$, *and the trial segments for all* $0 \le \alpha \le \alpha_0$, *for some fixed* $\alpha_0 > 0$, *lie in a common neighbourhood where the first and second derivatives of* $f, \boldsymbol{g}$ *are bounded;*
- $m_B \boldsymbol{I} \preceq \boldsymbol{B}^{(k)} \preceq M_B \boldsymbol{I}$ *for fixed* $0 < m_B \le M_B < \infty$; *and*
- *the subproblem multipliers are bounded and satisfy* $\rho - \left\|\widehat{\boldsymbol{\lambda}}^{(k+1)}\right\|_\infty \ge \eta$ *for a fixed* $\eta > 0$.

Then $\boldsymbol{d}^{(k)} \to \boldsymbol{0}$, $\boldsymbol{g}^{(k)} \to \boldsymbol{0}$, *and* $\nabla f^{(k)} + \boldsymbol{A}^{(k)} \widehat{\boldsymbol{\lambda}}^{(k+1)} \to \boldsymbol{0}$. *Every accumulation point of* $(\boldsymbol{x}^{(k)}, \widehat{\boldsymbol{\lambda}}^{(k+1)})$ *is a KKT pair.*

Proof. The descent bound in Lemma 16.5 gives

$$-D_k \ge m_B \left\|\boldsymbol{d}^{(k)}\right\|^2 + \eta \left\|\boldsymbol{g}^{(k)}\right\|_1 .$$

Subproblem stationarity and these bounds also imply bounded directions, since $\boldsymbol{B}^{(k)}\boldsymbol{d}^{(k)} = -\nabla f^{(k)} - \boldsymbol{A}^{(k)}\widehat{\boldsymbol{\lambda}}^{(k+1)}$. Let $\alpha_0 > 0$ keep all trial segments in a common neighbourhood with Taylor remainder constant $C > 0$. The Armijo test is then guaranteed for $\alpha \le \min\{1, \alpha_0, (1-\sigma)m_B/C\}$, so backtracking accepts step lengths bounded below by β times this positive constant. Summing the Armijo decreases and using a lower bound for ϕ_ρ on the compact region yields

$$\sum_k \left(m_B \left\|\boldsymbol{d}^{(k)}\right\|^2 + \eta \left\|\boldsymbol{g}^{(k)}\right\|_1\right) < \infty.$$

Hence $\boldsymbol{d}^{(k)} \to \boldsymbol{0}$ and $\boldsymbol{g}^{(k)} \to \boldsymbol{0}$. Subproblem stationarity gives

$$\nabla f^{(k)} + \boldsymbol{A}^{(k)}\widehat{\boldsymbol{\lambda}}^{(k+1)} = -\boldsymbol{B}^{(k)}\boldsymbol{d}^{(k)} \to \boldsymbol{0}.$$

Any convergent subsequence of primal points and their bounded subproblem multipliers therefore limits to a KKT pair. □

This calculation explains how merit decrease can contribute to a KKT convergence proof, and why decrease of function values alone is insufficient. It also shows where approximate inner solves would enter: their feasibility and stationarity errors must be controlled in these estimates. The basic fixed-parameter code does not enforce all of the uniform assumptions above.

16.8.3 Merit-parameter updates beyond the basic algorithm

If a fixed ρ fails the descent check, one possible extension is to increase it after the model solve. For example, under the full positive curvature condition, choose $\eta > 0$ and an initial $\rho^{(-1)} > 0$. For $k = 0, 1, \ldots$, the rule

$$\rho^{(k)} = \max\{\rho^{(k-1)}, \left\|\widehat{\boldsymbol{\lambda}}^{(k+1)}\right\|_\infty + \eta\}$$

makes the multiplier bound hold for every nonzero computed step. The parameter must then remain fixed throughout that iteration's line search. Neither the direction nor the subproblem multiplier needs to be recomputed, because the model equations do not involve ρ.

One can also inspect the directional derivative directly. At an infeasible point, $\|\boldsymbol{g}\|_1 > 0$, so choosing $\rho > \nabla f^{\mathsf{T}}\boldsymbol{d}/\|\boldsymbol{g}\|_1$ makes it negative, regardless of the sign of the numerator. At a feasible point the penalty contributes no first-order change along a linearized-feasible step, so increasing ρ cannot repair a non-descent tangent direction. Its curvature or the direction itself must be reconsidered.

These observations explain how merit safeguards can be designed. They do not by themselves prove convergence when the merit function changes with k: the summation argument above used a fixed function and uniform bounds. A full analysis must control the parameter sequence as well. The supplied LN/SQP comparison keeps ρ and the optional shift fixed throughout each run.

16.8.4 Inconsistent linearized constraints

A mixed linearized model can be infeasible although the nonlinear problem is feasible. One may relax the linearization using nonnegative elastic variables:

$$\begin{aligned} \min_{\boldsymbol{d},\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}} \quad & \nabla f^{\mathsf{T}}\boldsymbol{d} + \tfrac{1}{2}\boldsymbol{d}^{\mathsf{T}}\boldsymbol{B}\boldsymbol{d} + \rho(\mathbf{1}^{\mathsf{T}}\boldsymbol{u} + \mathbf{1}^{\mathsf{T}}\boldsymbol{v} + \mathbf{1}^{\mathsf{T}}\boldsymbol{w}) \\ \text{subject to} \quad & \boldsymbol{g} + \boldsymbol{A}^{\mathsf{T}}\boldsymbol{d} = \boldsymbol{u} - \boldsymbol{v}, \qquad \boldsymbol{h} + \boldsymbol{C}^{\mathsf{T}}\boldsymbol{d} \le \boldsymbol{w}, \qquad \boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \ge \mathbf{0}. \end{aligned}$$

The variables $\boldsymbol{u}-\boldsymbol{v}$ relax equalities and $\boldsymbol{w}$ relaxes inequalities, while their penalty discourages unnecessary relaxation. Alternatively, a separate restoration phase can seek a point with a more useful linearization. Infeasibility of one quadratic subproblem alone does not certify nonlinear infeasibility.

16.9 Exercises

Exercise 16.1. For

$$\min_{\boldsymbol{x}\in\mathbb{R}^2} \tfrac{1}{2}[(x_1-2)^2 + (x_2+1)^2] \quad \text{subject to} \quad x_1 + x_2^2 - 1 = 0,$$

derive the first LN step from $\boldsymbol{x}^{(0)} = (1,0)^{\mathsf{T}}$, $\lambda^{(0)} = 0$. Write the equivalent SQP system, identify its new multiplier, and compare linearized with actual feasibility at the new point.

Exercise 16.2. For a quadratic objective and an affine equality $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{x} = \boldsymbol{b}$, show that one nonsingular full LN step reaches the KKT pair from every initial pair. State the additional curvature condition that makes this pair the minimizer.

Exercise 16.3. For the direction in Equation (16.18), replace the merit parameter by an arbitrary $\rho > 0$. Determine the exact interval of positive step lengths giving strict merit decrease and the interval satisfying Armijo with $0 < \sigma < 1$.

Exercise 16.4. Construct an indefinite symmetric 3×3 matrix $\boldsymbol{B}$ and a full-rank 3×2 matrix $\boldsymbol{A}$ for which $\boldsymbol{Z}^{\mathsf{T}}\boldsymbol{B}\boldsymbol{Z} \succ 0$. Choose a nonzero linear objective and a linearized equality, solve the resulting quadratic subproblem, and verify the nonsingularity.

Exercise 16.5 (Supplementary)**.** Give a feasible nonlinear inequality problem with an infeasible linearization. Introduce an elastic variable and interpret its penalty.